

Session 9: Transactions & Error HandlingTrainer

Coverage:COMMIT, ROLLBACK, SAVEPOINT.Triggers: use case “Auto update user’s balance after expense entry.”AI Integration: Use ChatGPT to simulate test triggers and rollback scenarios.

Question1: Create a SQL script to insert a new order into an Orders table, then use COMMIT to save the transaction and verify that the new order persists after reconnecting to the database.

Ans:

USE mydb;

CREATE TABLE IF NOT EXISTS

**Orders (order_id INT PRIMARY
KEY, customer_name**

VARCHAR(100), order_amount

DECIMAL(10,2)

);

START TRANSACTION;

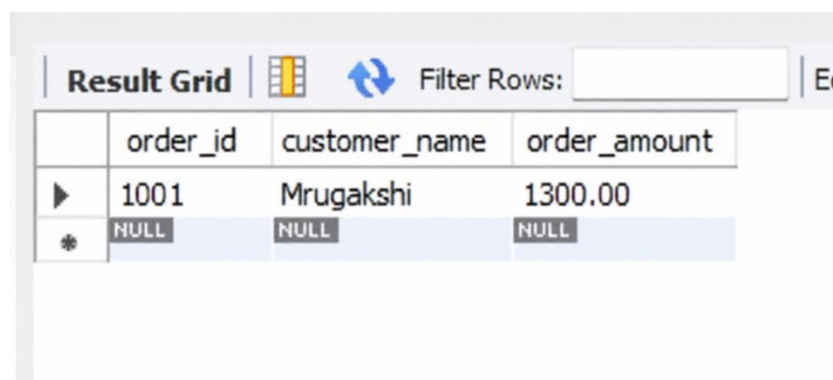
INSERT INTO Orders (order_id, customer_name, order_amount)

VALUES (1001, 'Mrugakshi', 1300.00);

COMMIT;

SELECT * FROM Orders

WHERE order_id = 1001;



The screenshot shows a 'Result Grid' window with a 'Filter Rows' input field. The grid contains three columns: 'order_id', 'customer_name', and 'order_amount'. The first row shows the inserted data: '1001', 'Mrugakshi', and '1300.00'. A second row is highlighted with a blue background and contains 'NULL' values for all three columns, indicating a new, empty record is ready for insertion. A small asterisk icon is visible in the first column of this row.

	order_id	customer_name	order_amount
▶	1001	Mrugakshi	1300.00
*	NULL	NULL	NULL

Question2: Simulate a Zomato-style food order process: insert two new items into an OrderItems table, then use ROLLBACK to undo the changes before committing. Check that no new items remain in the table after rollback.

Ans:

USE mydb;

DROP TABLE IF EXISTS

OrderItems; CREATE TABLE

OrderItems (item_id INT

PRIMARY KEY, order_id INT,

item_name VARCHAR(100),

quantity INT,

price DECIMAL(10,2)

) ENGINE=InnoDB;

START TRANSACTION;

INSERT INTO OrderItems

(item_id, order_id, item_name, quantity, price)

VALUES

(101, 500, 'Meggie', 1, 250.00),

(102, 500, 'Fries', 2, 180.00);

SELECT * FROM OrderItems;

ROLLBACK;

SELECT * FROM OrderItems;

Result Grid					
Filter Rows:					
	item_id	order_id	item_name	quantity	price
▶	101	500	Meggie	1	250.00
	102	500	Fries	2	180.00
*	NULL	NULL	NULL	NULL	NULL

Question3: In a Flipkart-like shopping cart scenario, use SAVEPOINT to mark a point after adding a product to the Cart table, add another product, then use ROLLBACK TO SAVEPOINT to undo only the last addition. Show the final contents of the Cart table.

Ans:

USE mydb;

DROP TABLE IF EXISTS Cart;

CREATE TABLE Cart (

cart_id INT PRIMARY KEY,

product_name VARCHAR(100),

quantity INT,

price DECIMAL(10,2)

) ENGINE=InnoDB;

START TRANSACTION;

INSERT INTO Cart (cart_id, product_name, quantity, price)

VALUES (1, 'Speaker', 1, 25000.00);

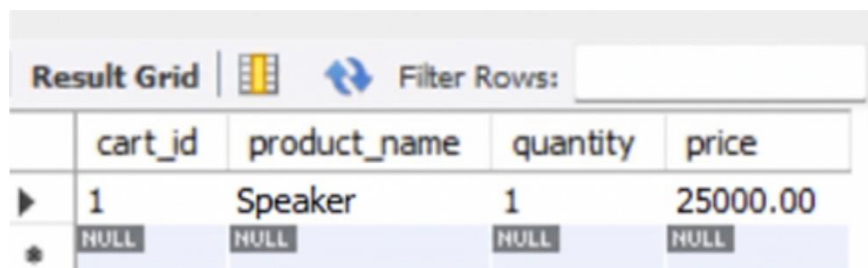
SAVEPOINT product_added;

INSERT INTO Cart (cart_id, product_name, quantity, price)

VALUES (2, 'Mobile', 1, 20000.00);

ROLLBACK TO SAVEPOINT product_added;

SELECT * FROM Cart;



The screenshot shows a database interface with a 'Result Grid' tab. It displays the contents of the 'Cart' table. The table has four columns: 'cart_id', 'product_name', 'quantity', and 'price'. There is one data row with values (1, 'Speaker', 1, 25000.00). Below the data row is a row with all 'NULL' values, indicating that the second product ('Mobile') was successfully rolled back. The interface also includes a 'Filter Rows' search bar and a refresh icon.

	cart_id	product_name	quantity	price
▶	1	Speaker	1	25000.00
✱	NULL	NULL	NULL	NULL

Question4: Write a trigger on a Wallet table that automatically deducts the purchase amount from the user's balance whenever a new transaction is inserted, similar to how Paytm updates wallet balance after a payment.

Ans:

USE mydb;

DROP TABLE IF EXISTS Transactions;

DROP TABLE IF EXISTS

Wallet; CREATE TABLE

Wallet (user_id INT

PRIMARY KEY,

user_name VARCHAR(100),

balance DECIMAL(10,2)

);

CREATE TABLE Transactions (

transaction_id INT PRIMARY KEY,

user_id INT,

purchase_amount DECIMAL(10,2)

);

INSERT INTO Wallet VALUES

(1, 'mrugakshi', 13000.00);

DELIMITER //

CREATE TRIGGER deduct_wallet_balance

AFTER INSERT ON Transactions

FOR EACH ROW

BEGIN

UPDATE Wallet

SET balance = balance - NEW.purchase_amount

WHERE user_id = NEW.user_id;

```

END //

DELIMITER ;

INSERT INTO Transactions
VALUES (101, 1, 500.00);

SELECT * FROM Wallet;

```

	user_id	user_name	balance
▶	1	mrugakshi	12500.00
✱	NULL	NULL	NULL

Question5: Use ChatGPT to generate a SQL test scenario where a trigger incorrectly updates a user's balance after an expense entry. Paste the scenario and your corrected trigger code, explaining how you fixed the error.

Hint: Ask ChatGPT for a buggy trigger example and a test case that exposes the bug.

Ans:

```

USE mydb;

CREATE TABLE Wallet (
  user_id INT PRIMARY
  KEY,
  balance
  DECIMAL(10,2)
);

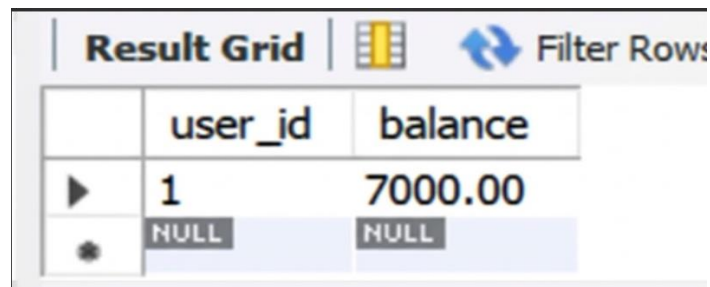
CREATE TABLE Expense (
  expense_id INT
  AUTO_INCREMENT PRIMARY KEY,
  user_id
  INT,
  amount DECIMAL(10,2)
);

INSERT INTO Wallet (user_id, balance)

```

VALUES (1, 7000);

SELECT * FROM Wallet;



	user_id	balance
▶	1	7000.00
●	NULL	NULL

Buggy Trigger:

```
DELIMITER //  
CREATE TRIGGER update_balance  
AFTER INSERT ON Expense  
FOR EACH ROW  
  
BEGIN  
    UPDATE Wallet  
    SET balance = balance + NEW.amount  
    WHERE user_id = NEW.user_id;  
END//  
DELIMITER ;  
INSERT INTO Expense (user_id, amount)  
VALUES (1, 1000);  
SELECT * FROM Wallet;
```



	user_id	balance
▶	1	8000.00
●	NULL	NULL

DROP TRIGGER IF EXISTS update_balance;

DELIMITER //

CREATE TRIGGER update_balance

AFTER INSERT ON Expense

FOR EACH ROW

BEGIN

UPDATE Wallet

SET balance = balance - NEW.amount

WHERE user_id = NEW.user_id;

END//

DELIMITER ;

UPDATE Wallet

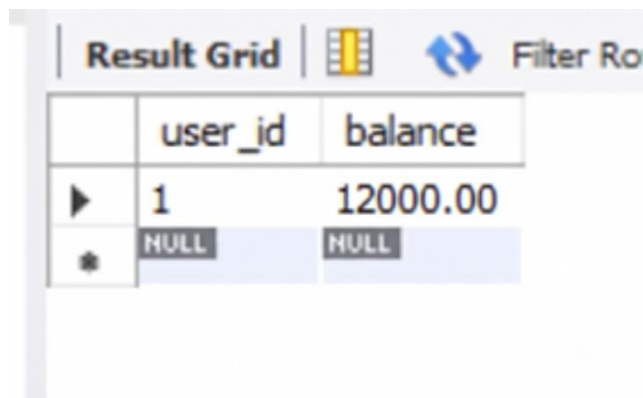
SET balance = 13000

WHERE user_id = 1;

INSERT INTO Expense (user_id, amount)

VALUES (1, 1000);

SELECT * FROM Wallet;



The screenshot shows a 'Result Grid' window with a toolbar containing icons for a grid, a refresh button, and a 'Filter Rows' label. The grid displays the results of the SQL query 'SELECT * FROM Wallet;'. It has two columns: 'user_id' and 'balance'. The first row shows '1' for user_id and '12000.00' for balance. The second row shows 'NULL' for both user_id and balance, indicating an empty record.

	user_id	balance
▶	1	12000.00
*	NULL	NULL